

EEE 316 (January 2024)

Power Electronics Laboratory

Final Project Report

Section: A2 Group: 05

Project Title: Auto Cut Off Battery Charger and Smart Battery Level Indicator

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1 Abstract

The project aims to develop a multifunctional battery charger equipped with an adjustable DC output, employing a Buck converter along with an automatic cutoff mechanism. This charger is capable of charging a variety of battery types such as lead-acid and lithium-ion batteries. The system features an AC-DC converter, specifically an uncontrolled full-wave rectifier paired with an LC filter which converts alternating current input into a stable, ripple-free output voltage. The Buck converter, managed by the IC SG3524, modifies the output voltage by altering the switching frequency and duty cycle, which can be precisely adjusted using a potentiometer to meet the specific requirements of different batteries. Furthermore, the charger is designed with an automatic cutoff and charging system that ceases operation once the battery reaches full charge. This feature is realized through the use of relay, a microcontroller, and BJTs. To enhance safety, the design incorporates protective measures against over-voltage, over-current, and reverse polarity, thereby preventing overcharging and ensuring the system's integrity in the event of faults.

2 Introduction

The increasing reliance on rechargeable batteries in various applications highlights the need for efficient and adaptable charging systems. This project focuses on developing a multifunctional battery charger capable of handling different battery types, including lead-acid and lithium-ion, by providing an adjustable DC output regulated through a Buck converter.

Using the IC SG3524, the charger ensures precise voltage and current control to meet specific charging requirements. It features an AC-DC conversion stage for stable output and an automatic cutoff mechanism to halt charging when the battery reaches full capacity.

To enhance safety and reliability, the charger is equipped with protections against over-voltage, over-current, and reverse polarity. This report details the design, functionality, and testing of the charger, emphasizing its flexibility and secure operation.

3 Design

3.1 Problem Formulation

3.1.1 Identification of Scope

The multifunctional battery charger developed in this project is designed to address the charging requirements of various battery types, particularly lead-acid and lithium-ion batteries, by offering adjustable DC output voltage and current. The scope of this project encompasses:

1. **Battery Compatibility:** The charger supports multiple battery chemistries with customizable charging parameters to ensure optimal performance and longevity.
2. **Voltage Regulation:** A Buck converter, controlled by the IC SG3524, provides precise voltage and current adjustments to suit different battery specifications.
3. **Automatic Cutoff:** The system includes an automatic cutoff mechanism to stop charging when the battery is fully charged, protecting against overcharging.
4. **Safety Features:** Comprehensive protections, including over-voltage, over-current, and reverse polarity safeguards, enhance system reliability and prevent damage.
5. **User Control:** A user-friendly interface, including a potentiometer for fine-tuning output voltage, offers flexibility in operation.
6. **Versatility:** The charger is applicable in various domains such as renewable energy storage, automotive systems, and portable electronic devices.

This scope defines the operational boundaries and highlights the functionalities that make the charger a versatile, safe, and reliable solution for modern battery charging needs.

3.1.2 Literature Review

Battery charging systems have evolved to meet the demands for efficiency, safety, and compatibility with diverse battery types. Lead-acid and lithium-ion batteries, for instance, require distinct charging profiles—multi-stage charging for lead-acid and CC-CV modes for lithium-ion—highlighting the need for adaptable chargers.

Buck converters are widely recognized for their efficiency in voltage regulation. The IC SG3524, frequently utilized in power electronics, provides precise control of voltage and current, making it suitable for variable-output charging systems.

Automatic cutoff mechanisms, often implemented with microcontrollers and relays, ensure charging stops when batteries are fully charged, preventing overcharging and conserving energy. Safety features like over-voltage, over-current, and reverse polarity protection further enhance system reliability, as documented in power system protection research.

Finally, multi-battery compatibility is increasingly valued in sectors such as renewable energy and electric vehicles. Adjustable output parameters allow chargers to serve a range of battery chemistries, ensuring flexibility and broad usability.

3.1.3 Formulation of Problem

The challenge lies in developing a multifunctional battery charger that:

1. **Provides Adjustable Output:** Offers precise voltage and current regulation to cater to the unique requirements of different battery types.
2. **Ensures Safety:** Incorporates protections against over-voltage, over-current, and reverse polarity to safeguard both the charger and the battery.
3. **Prevents Overcharging:** Features an automatic cutoff mechanism to stop charging once the battery is fully charged.
4. **Maintains Reliability:** Delivers stable performance while being easy to operate and versatile enough for multiple applications.

This project addresses these issues by designing and implementing a charger that combines an AC-DC conversion stage, a Buck converter, and advanced safety mechanisms, providing a comprehensive and reliable solution for modern battery charging needs.

3.1.4 Analysis

The battery charger combines efficiency, versatility, and safety to meet diverse charging needs:

1. Design Efficiency

- The Buck converter, controlled by the IC SG3524, ensures precise voltage and current adjustments, supporting various battery types.
- An AC-DC conversion stage with a full-wave rectifier and LC filter provides a stable DC output.

2. Functionality

- Adjustable output via a potentiometer allows fine-tuning for different batteries.
- An automatic cutoff system, managed by a relay, microcontroller, and BJTs, prevents overcharging and conserves energy.

3. Safety Measures

- Over-voltage, over-current, and reverse polarity protections safeguard the system and batteries.

4. Usability and Applications

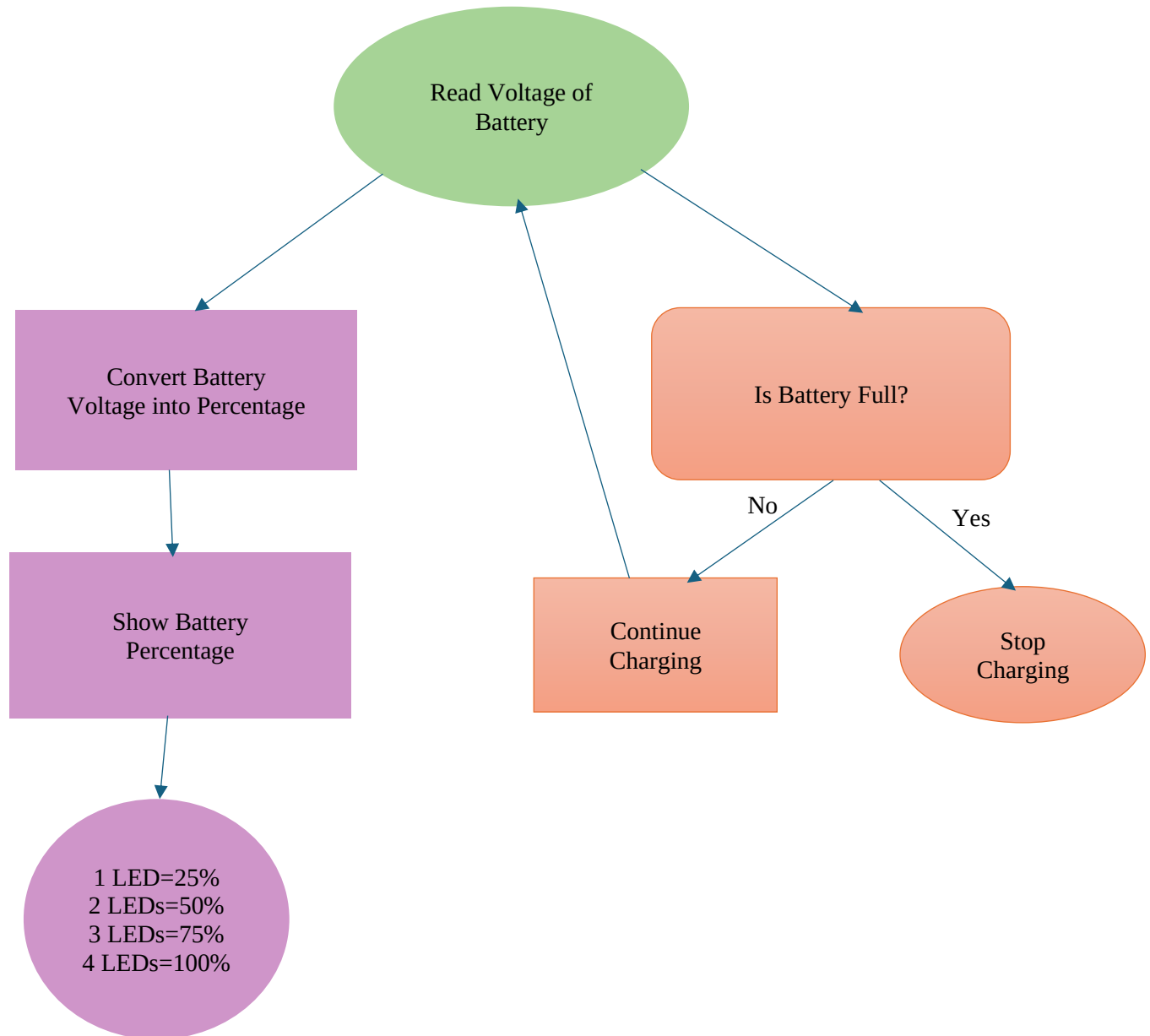
- The charger supports multiple applications, including automotive systems and renewable energy storage, with a user-friendly design.

5. Performance and Reliability

- High efficiency and stability are ensured by reliable components and thorough testing.

This analysis highlights the charger's adaptability, safety, and effectiveness for modern battery charging requirements.

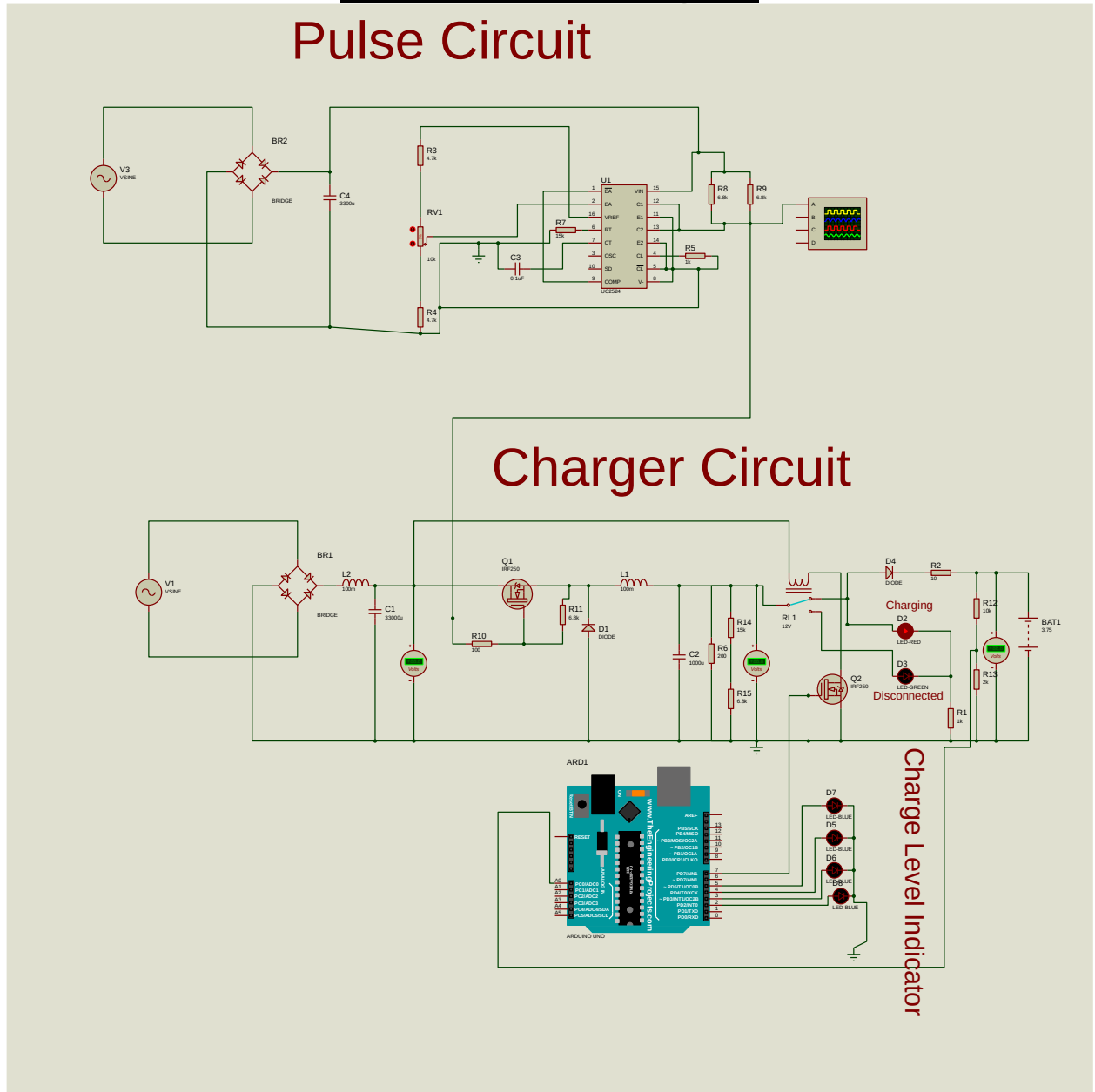
3.2 Design Method



3.3 Circuit Diagram

3.3.1 Proteus Circuit Diagram

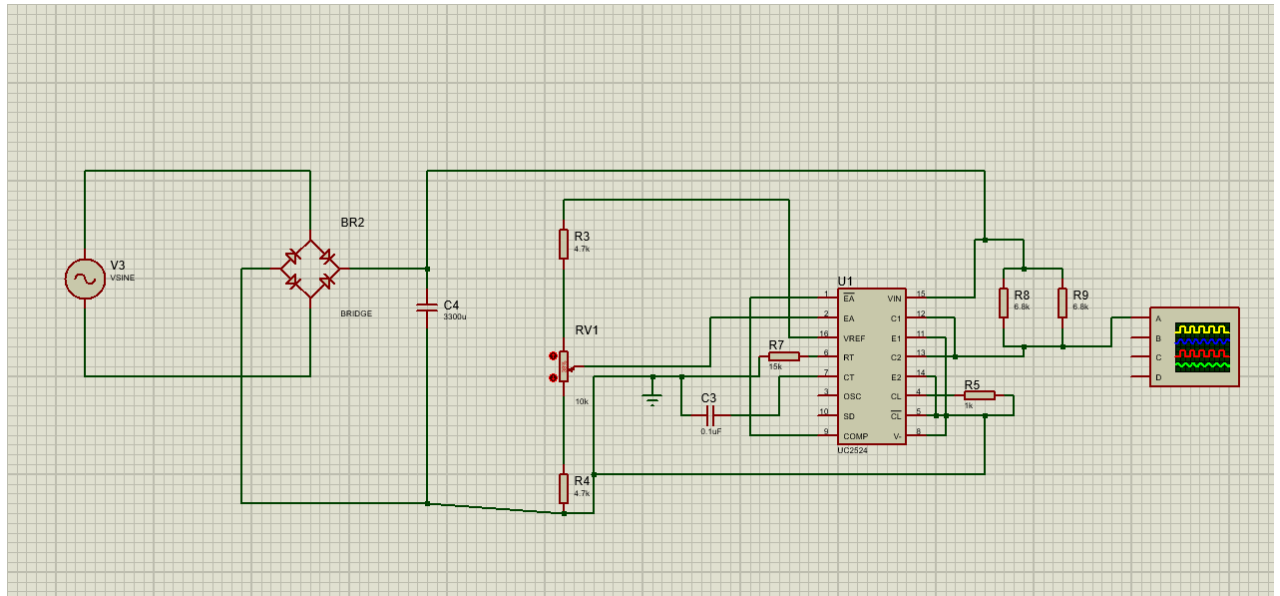
Pulse Circuit



Charger Circuit Overview:

This circuit is designed for charging batteries with charge level monitoring and indicators. It converts AC to DC, regulates the charging process, and uses an Arduino to display the battery's charge status through LEDs. It ensures efficient and safe charging while providing real-time feedback.

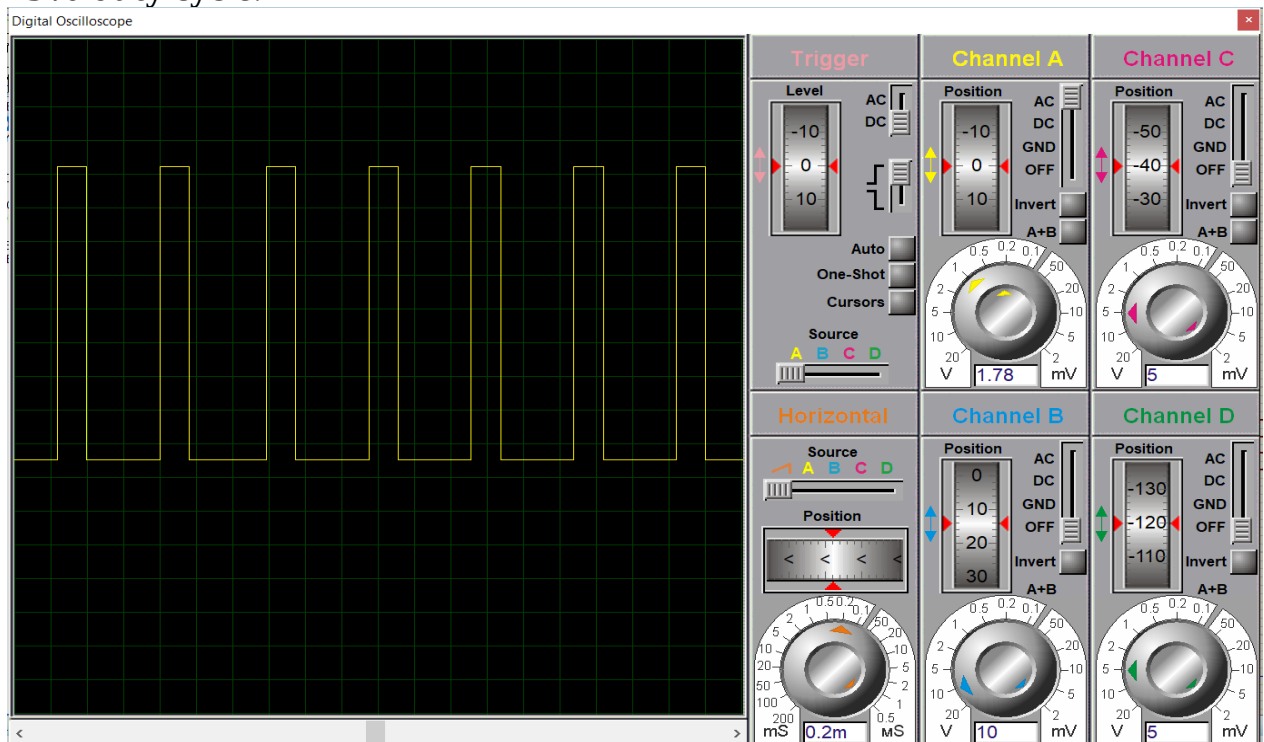
3.3.2 Proteus circuit simulation for Pulse Circuit:



Circuit Description: The circuit is a power supply system using the UC3524 PWM controller IC. It converts AC input to a DC voltage and generates PWM signals for controlling switching devices. The UC3524 regulates the output based on feedback, with adjustable parameters for frequency and duty cycle.

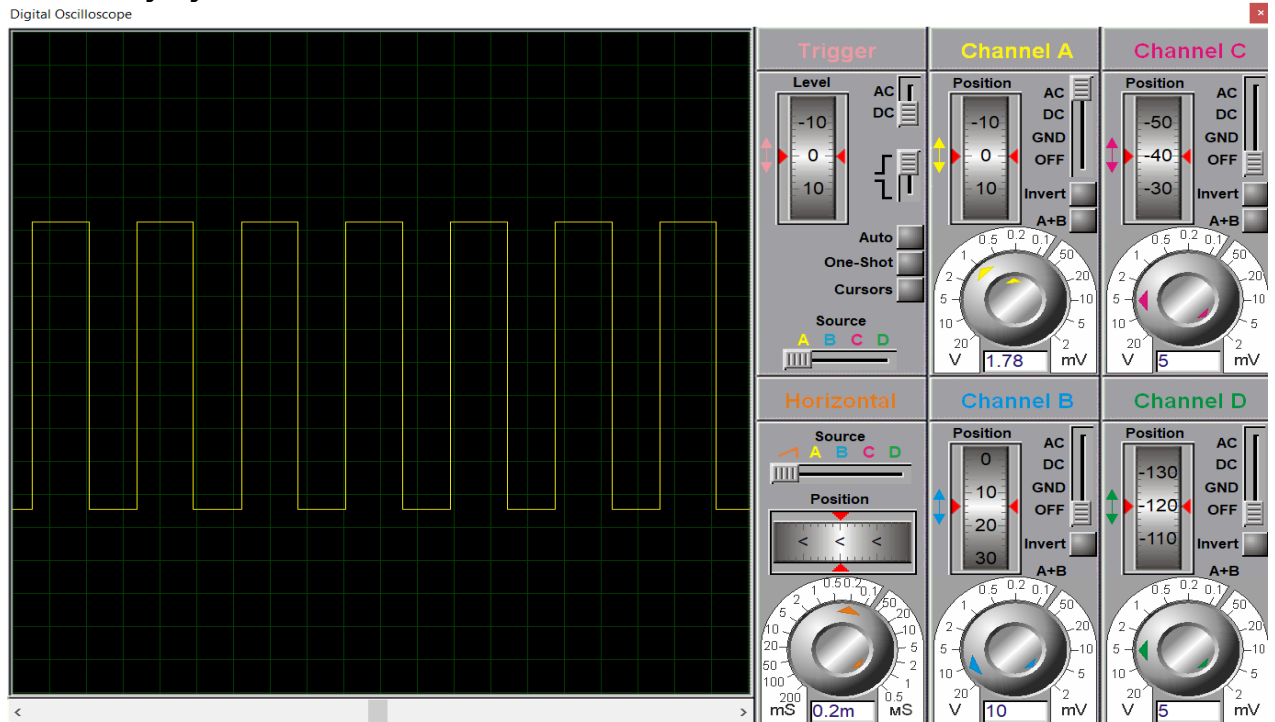
Pulse signal of proteus:

25% duty cycle:



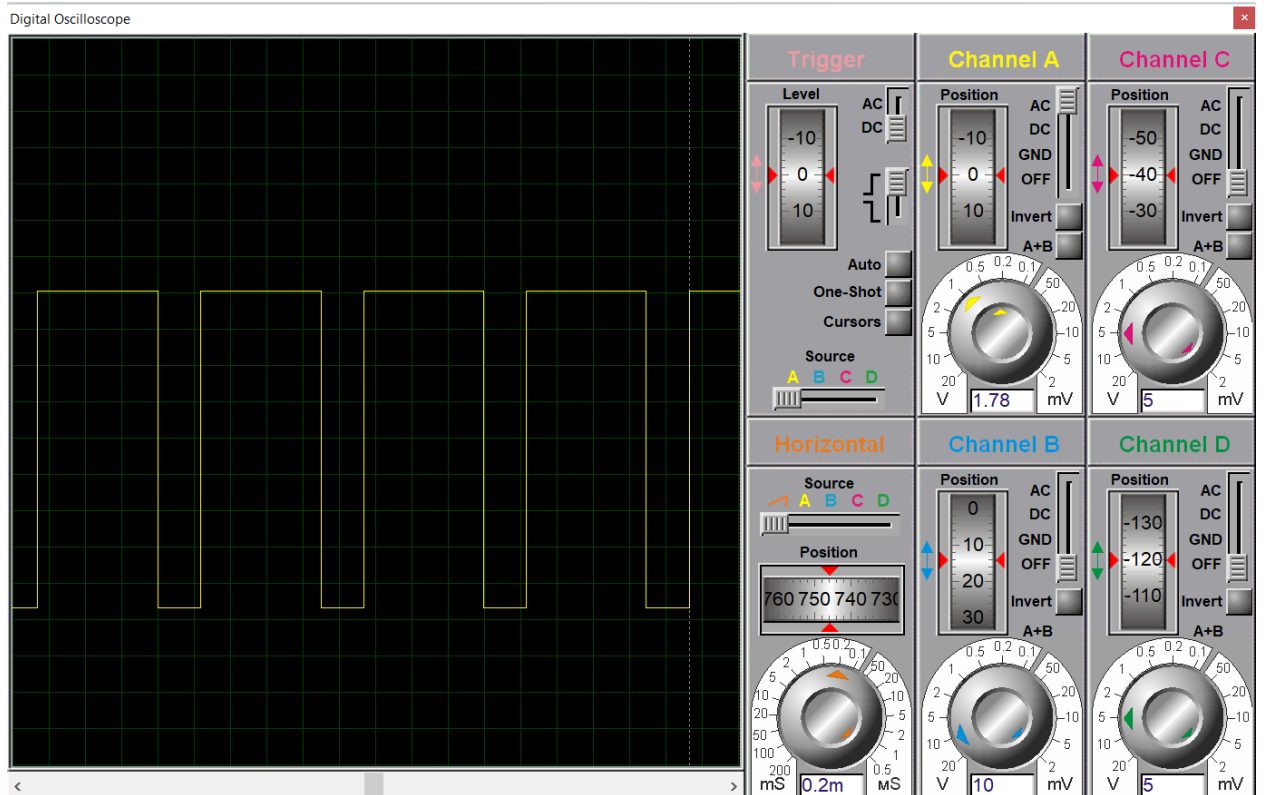
50% duty cycle:

Digital Oscilloscope

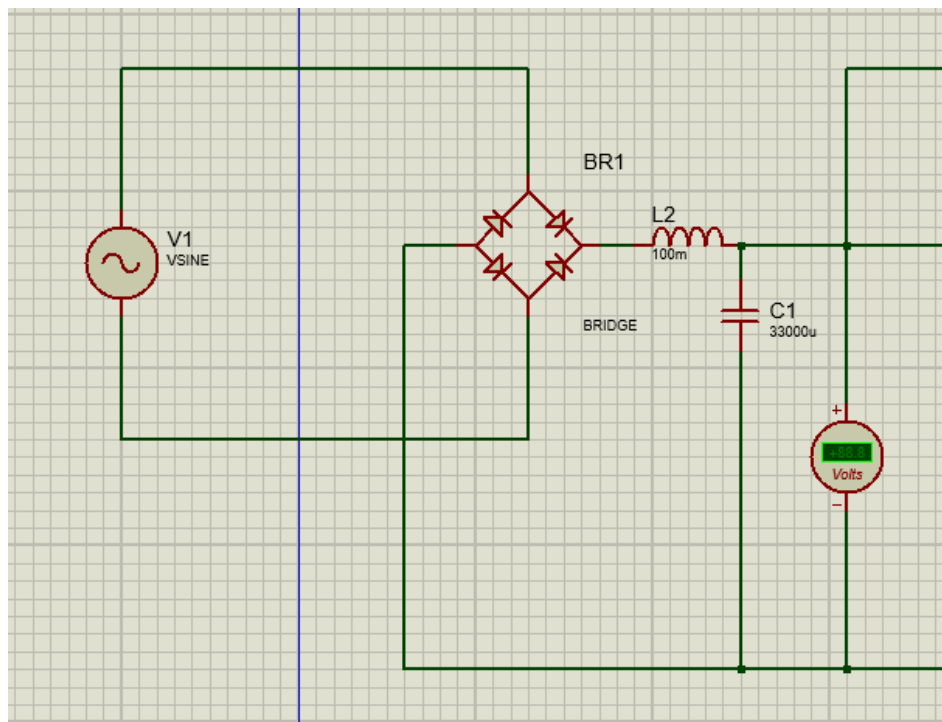


75 % duty cycle:

Digital Oscilloscope



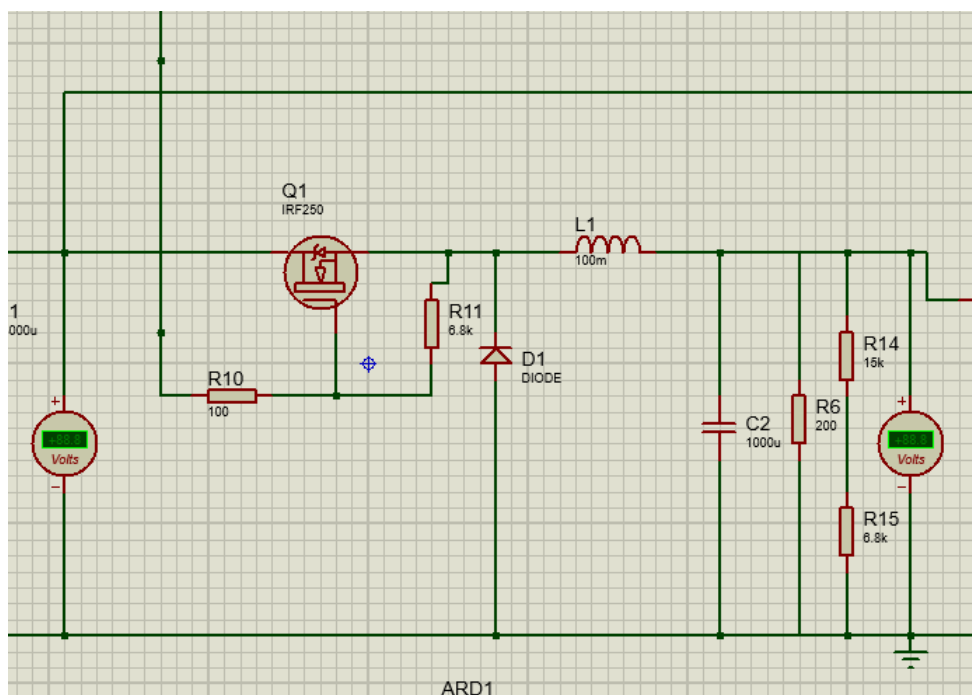
3.3.3 Rectifier Circuit:



Circuit Description:

This circuit rectifies AC signal and converts it into a DC signal which has ripples. To mitigate the ripples a LC filter is used.

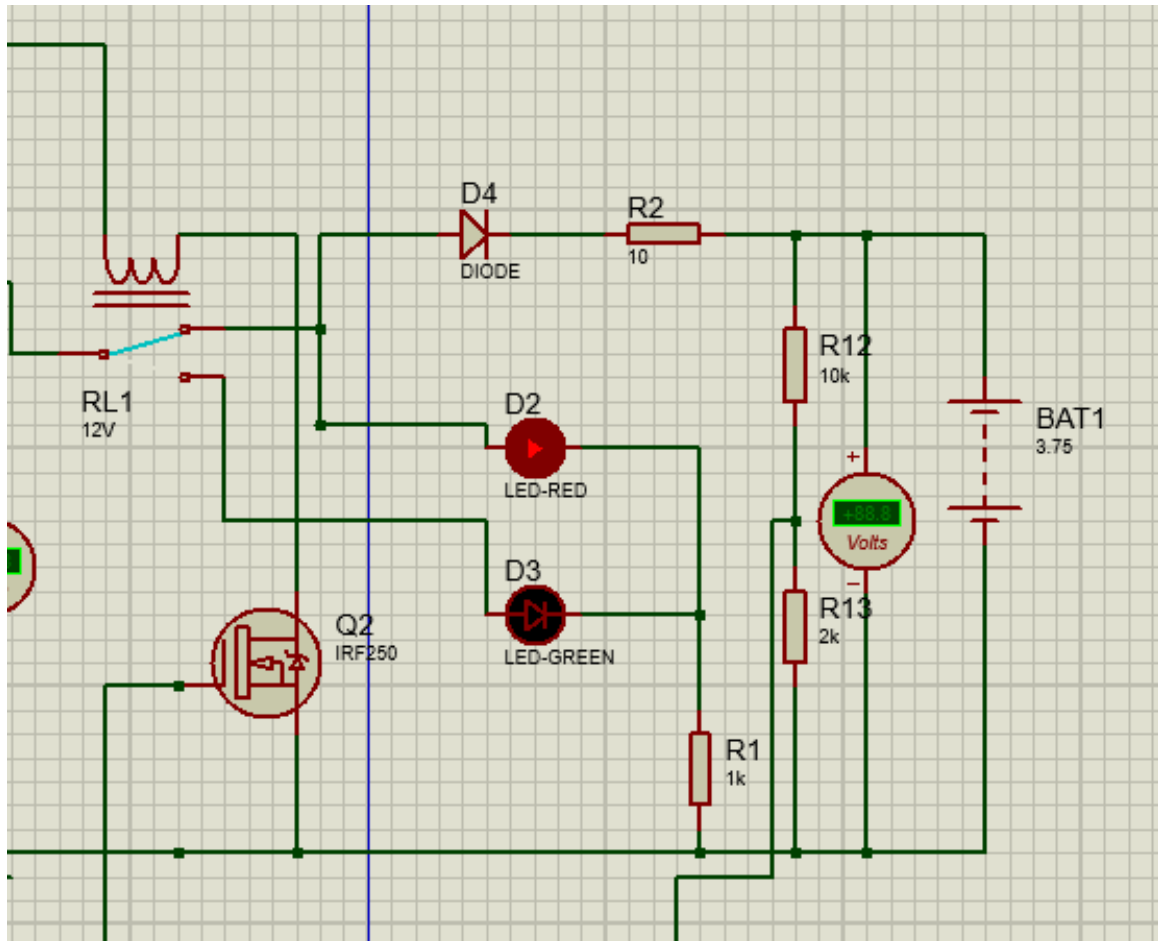
3.3.4 Buck Converter Circuit:



Circuit Description:

This circuit is a stepdown DC-DC converter. Depending on duty cycle set by the pulse circuit, the output voltage varies from 0 to input voltage level.

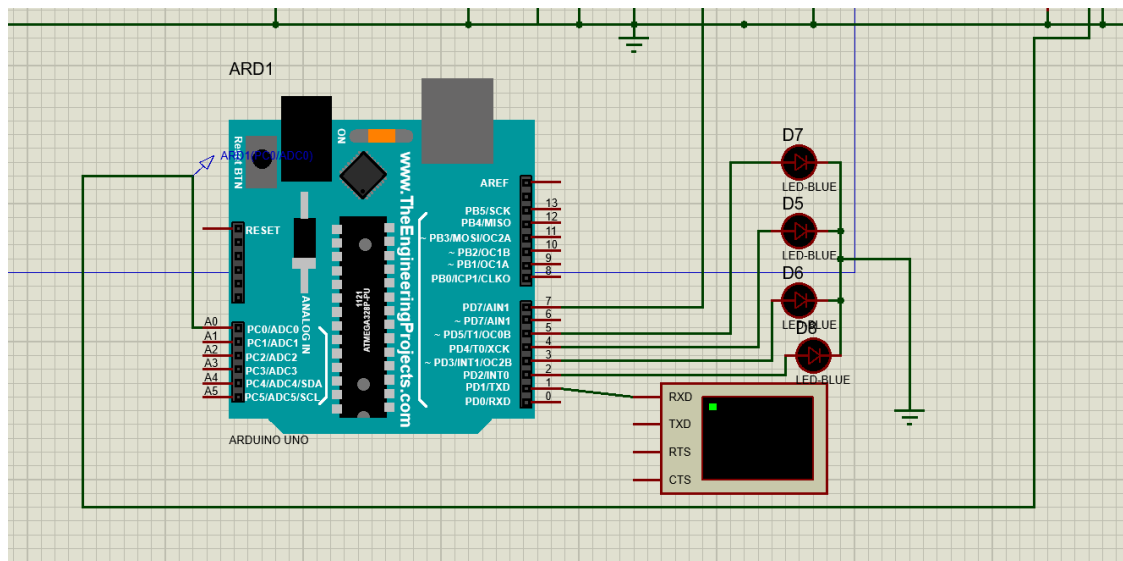
3.3.5 Auto Cutoff Circuit:



Circuit Description:

This circuit has 2 LEDs, Red and Green. Green indicates not charging mode and Red indicates the charging mode. This circuit contains a relay which can switch between two modes. The circuit cuts off charging when the battery is full.

3.3.6 Level Indicator Circuit:



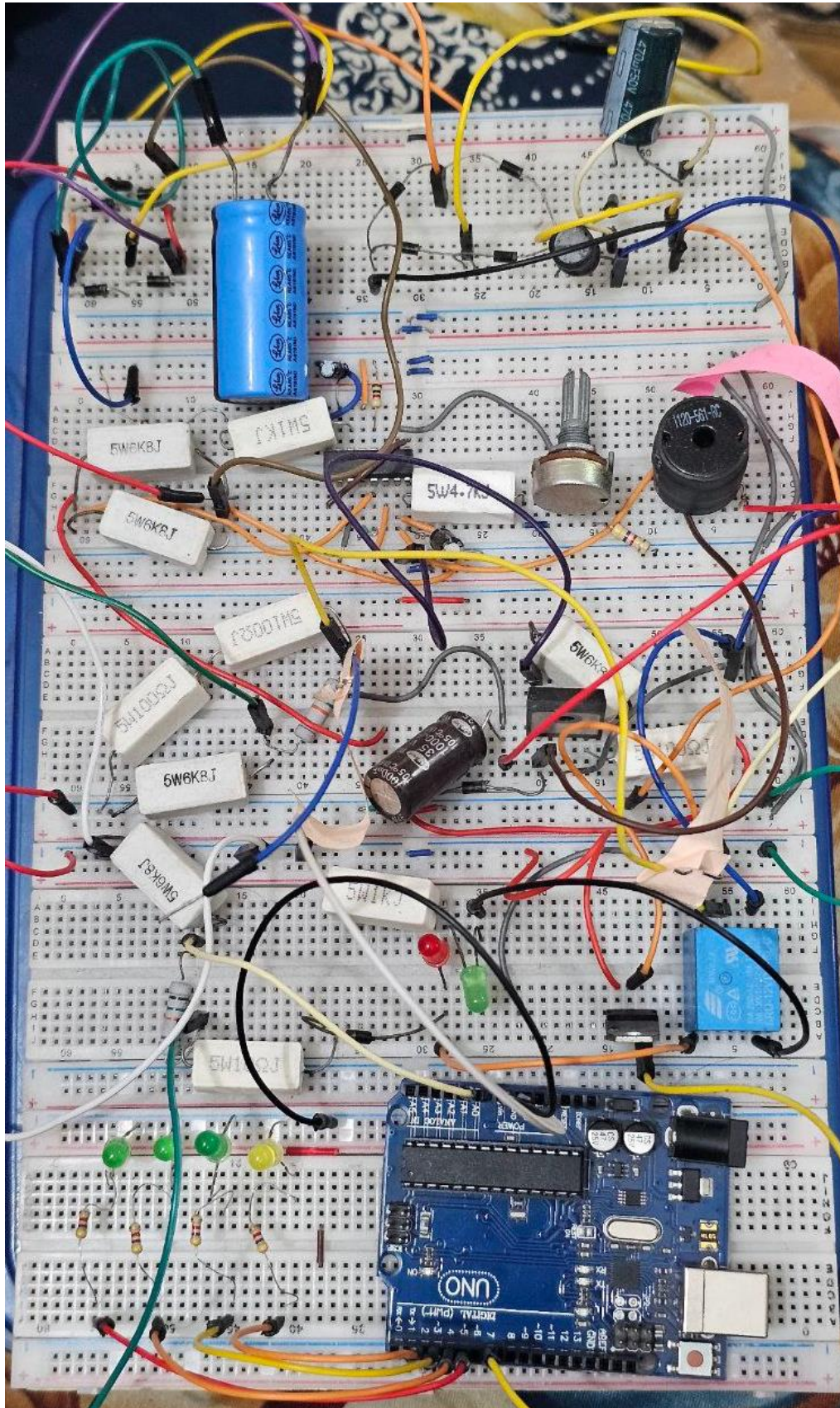
Circuit Description:

This Circuit show the current battery level with 4 LED. A microcontroller was used to accurately measure and indicate the current battery level.

3.4 Hardware Design

Components:

- Step down transformer (220V/12V) 3000mA
- Silicon Diodes (1N4001)
- Capacitors (1000uF,3300uF, 220uF,0.01uF)
- Inductors (100mH)
- Resistors (10, 100, 220,4.7k,6.8k,15k,2k,1k, 10k)
- Lithium Ion Battery - 2 pieces (each rated 3.7V)
- SG3524
- 10k potentiometer
- Arduino UNO
- Voltage measurement IC
- BJT BD-243
- Jumper Wires



3.5 Full Source Code of Firmware

```
autoCutOff3.ino
1 // defining Pin
2 const int analogPin = A0; // Analog pin connected to the voltage divider
3 const int bjtPin = 7; // Digital pin controlling the BJT
4 const int ledPins[] = {2, 3, 4, 5}; // Pins for 4 LEDs representing 25% each
5
6 // Voltage divider resistors
7 const float R1 = 15000.0; // Resistor R1 in ohms
8 const float R2 = 6800.0; // Resistor R2 in ohms
9
10 // Reference voltage and battery limits
11 const float vRef = 5.0; // Arduino reference voltage (5V)
12 const float fullVoltage = 4.2; // Full-charge voltage for the 3.7V battery
13 const float emptyVoltage = 1.0; // Minimum voltage for 0% battery
14
15 void setup() {
16     pinMode(bjtPin, OUTPUT);
17     for (int i = 0; i < 4; i++) {
18         pinMode(ledPins[i], OUTPUT);
19     }
20     digitalWrite(bjtPin, HIGH); // Start charging
21     Serial.begin(9600);
22 }
23
24 void loop() {
25     // Reading the battery voltage
26     int adcValue = analogRead(analogPin);
27     float batteryVoltage = (adcValue * vRef / 1023.0) * 3.20588; // Updated ratio
28
29     // Calculate battery percentage
30     float batteryPercentage = mapBatteryPercentage(batteryVoltage);
31
32     // Display battery percentage on LEDs
33     lightUpLEDs(batteryPercentage);
34
35     // Control charging based on battery voltage
36     if (batteryVoltage >= fullVoltage) {
37         digitalWrite(bjtPin, LOW); // Stop charging
38         Serial.println("Battery Full. Charging stopped.");
39     } else {
40         digitalWrite(bjtPin, HIGH); // Continue charging
41         Serial.println("Charging...");
42     }
43
44     // Print battery status to Serial Monitor
45     Serial.print("Battery Voltage: ");
46     Serial.print(batteryVoltage);
47     Serial.print(" V, Battery Percentage: ");
48     Serial.print(batteryPercentage);
49     Serial.println(" %");
50
51     delay(1000); // Update every second
52 }
53
54 // Function to map battery voltage to percentage
55 float mapBatteryPercentage(float voltage) {
56     if (voltage >= fullVoltage) return 100.0;
57     if (voltage <= emptyVoltage) return 0.0;
58     return (voltage - emptyVoltage) / (fullVoltage - emptyVoltage) * 100.0;
59 }
60
```

```

61 // Function to light up LEDs based on battery percentage
62 void lightUpLEDs(float percentage) {
63     int numLEDsOn = (int)(percentage / 25.0); // Determine how many LEDs should be ON
64     for (int i = 0; i < 4; i++) {
65         if (i < numLEDsOn) {
66             digitalWrite(ledPins[i], HIGH); // Turn ON LEDs
67         } else {
68             digitalWrite(ledPins[i], LOW); // Turn OFF LEDs
69         }
70     }
71 }
72

```

4 Implementation

4.1 Description

Switching Pulse Generation Circuit:

Switching pulses are generated by using SG3524 IC which is powered with a rectifier circuit. The frequency of the pulse is dependent on the values of R_T and C_T .

$$\text{Switching frequency, } f = \frac{1.3}{C_T R_T}$$

We used $R_T = 1\text{k}\Omega$ and $C_T = 0.1\mu\text{F}$. So, $f = 13\text{ kHz}$

Buck Converter Circuit:

$$L_{\min} = \frac{(1-K)R}{2f}$$

From the circuit, equivalent resistance $R \cong 200\Omega$.

So, for $k = 0.1$, $L_{\min} = 6.92\text{ mH}$

For $k = 0.9$, $L_{\min} = 0.7\text{ mH}$

We used a 5mH inductor in buck converter, which is shorter than the $L_{\min}$ value. But it is okay as we would not increase our duty cycle absolute maximum and we needed to increase the output current by the ferrite core inductor we used.

$$C_{\min} = \frac{1-k}{16Lf^2}$$

$L = 5\text{ mH}$.

So, for $k = 0.1$, $C_{\min} = 70\text{nF}$

For $k = 0.9$, $C_{\min} = 7\text{ nF}$.

We chose $1000\mu\text{F}$ capacitor, which is way above the limiting value.

Buck converter:

Buck converter is a step-down converter. It can be controlled by changing the duty cycle of the switching pulse. We have used IRF 250 MOSFET as switch and gate pulse is applied between the gate and source of the MOSFET.

$V_o = k \cdot V_{in}$. Here, k is the duty cycle.

Auto cutoff circuit and auto charging circuit:

Relay coil is supplied with voltage from the rectifier output and the coil is connected to the ground through a BJT. Again, the BJT is driven by a microcontroller. When the BJT is on, the relay coil gets current and SPDT pole changes its position from the NO (Normally Open) position to NC (Normally Close) position. NO position is connected with the green indicator which indicates the battery is not charging. NC is connected to the positive terminal of the battery through a series diode and 10-ohm series power resistor. The resistor limits the short circuit current and the diodes prevents the back flow of current from the battery.

Microcontroller:

We used a voltage divider from battery to feed the voltage into Arduino as the Arduino can read maximum of 5V. The `analogRead()` function converts the analog voltage into a Digital number between 0 to 1023. So, the actual battery voltage after converting it into analog voltage again and using the voltage divider:

$$V_{\text{battery}} = \frac{\text{analogRead}(A0) \times V_{\text{ref}}}{1023} \times \frac{R1 + R2}{R2}$$

Here, V_{ref} is the reference voltage of Arduino which is 5V.

5 Design Analysis and Evaluation

5.1 Novelty

The multifunctional battery charger offers innovative features that distinguish it from conventional systems:

1. **Adjustable Output:** Supports multiple battery types (lead-acid, lithium-ion) with customizable voltage and current settings.
2. **Automatic Cutoff:** Intelligent mechanism using a relay, microcontroller, and BJTs to prevent overcharging.
3. **Enhanced Safety:** Incorporates over-voltage, over-current, and reverse polarity protections for reliability.
4. **Advanced Control:** Utilizes the SG3524 IC for precise voltage and current regulation via the Buck converter.
5. **Versatile Applications:** Suitable for automotive, renewable energy, and portable device use with a user-friendly interface.

This design ensures adaptability, safety, and efficiency for modern battery charging needs.

5.2 Design Considerations

5.2.1 Considerations to public health and safety

- The charger incorporates safety features like over-voltage, over-current, and reverse polarity protections to prevent battery damage or hazardous incidents such as overheating and electrical short circuits.
- The automatic cutoff mechanism ensures that batteries are not overcharged, reducing risks of battery failure or accidents

5.2.2 Considerations to environment

- The charger's efficient design reduces energy waste by ensuring precise charging and automatic termination upon full charge, conserving power.
- By supporting rechargeable batteries like lithium-ion and lead-acid, the charger promotes sustainable practices, reducing reliance on disposable batteries and minimizing environmental waste.

5.2.3 Considerations to cultural and societal needs

- The charger addresses the growing societal need for adaptable charging solutions in renewable energy and portable device applications.
- Its user-friendly interface ensures accessibility to a wide audience, including individuals with minimal technical expertise

5.3 Investigations

The “Auto Cut Off Battery Charger and Smart Battery Level Indicator” is innovative technologies designed to enhance the efficiency and safety of charging and maintaining batteries. The “Auto Cut Off Battery Charger” automatically detects when a battery is fully charged, preventing overcharging and extending the lifespan of the battery. This feature is particularly useful for users who want to ensure that their batteries are charged safely without constant monitoring. On the other hand, the “Smart Battery Level Indicator” provides real-time, accurate information about the battery's charge status, allowing users to easily monitor the remaining power and make informed decisions about when to recharge. Together, these technologies represent significant advancements in battery management, offering both convenience and protection for a wide range of electronic devices and vehicles.

5.3.1 Design of Experiment

- **Voltage and Capacity:** The charger is compatible with the battery's voltage and capacity to ensure safe and effective charging.
- **Charging Current and Voltage Limits:** The charger delivers appropriate current and voltage to avoid overcharging or damaging the battery.
- **Battery Protection:** The charger is designed to handle battery-specific safety features, like overcharge protection, under-voltage cut-off, and temperature thresholds.

- **Overcharging Prevention:** The automatic cut-off mechanism is designed to stop charging when the battery reaches its full capacity, preventing overcharging.
- **End of Charge Detection:** The charger detects when charging current drops below a certain threshold (indicating full charge) and automatically stop the charging process.
- **Power Conversion Efficiency:** The charger is designed with high efficiency to minimize energy loss and reduce heat generation.
- **LED indicator:** The charger provides a clear and easy-to-read indicator through LED lights.

5.3.2 Data Collection

Output of Rectifier: 16.5V DC with minor ripples.

Output of Buck Converter: 4.2V DC with negligible ripples

Output of Battery Charging End: 4V DC with negligible ripples

5.3.3 Results and Analysis

The auto-cut-off battery charger worked well. The charging procedure consistently achieved the desired voltage, and the automatic cut-off mechanism effectively prevented overcharging, extending the battery's life. The observed charging times fell within the expected range for the battery type and capacity. An examination of the charging current and voltage profiles demonstrated a smooth and consistent charging operation.

The smart battery level indicator accurately revealed the battery's charge status in real time. The indicator efficiently communicated the battery's status with distinct visual and aural clues. The charger has demonstrated high efficiency in terms of power conversion, with energy losses minimization. The efficiency of the charger ensures minimal power wastage.

The charger integrates multiple safety mechanisms, including short-circuit protection, reverse polarity protection and overvoltage/under voltage protection. These features have been successfully validated during testing, with no incidents of damage occurring under extreme conditions.

The “Auto Cut-Off Battery Charger and Smart Battery Level Indicator” have successfully met all the design requirements, offering a safe, efficient, and user-friendly solution for battery charging and monitoring. The system’s high charging efficiency, safety features, and accurate battery level monitoring ensure optimal performance and longevity of batteries, while the user interface provides clear feedback. The integration of advanced health management and optional communication features enhances the user experience, making the system suitable for both basic and advanced users.

5.4 Limitations of Tools

• IC SG3524

- Limited to specific voltage and current ranges, restricting its use for high-power or advanced charging profiles.
- Lacks support for advanced control algorithms, reducing precision compared to modern systems.

• Uncontrolled Full-Wave Rectifier

- Relies on external filters to minimize ripple, which may be less effective with fluctuating AC supply.
- **Relay-Based Cutoff**
 - Slower switching compared to solid-state devices and subject to wear over time, reducing reliability.
- **LC Filter**
 - Efficiency depends on component quality; may not eliminate high-frequency noise entirely.
- **User Interface**
 - Manual adjustments via a potentiometer are less precise than digital interfaces, limiting ease of use.
- **Safety Features**
 - Basic protections lack advanced diagnostics or thermal management for comprehensive fault detection.
- **Budget Constraints**
 - Limited resources may restrict the use of higher-quality components or advanced technologies, potentially affecting performance and scalability.

5.5 Impact Assessment

5.5.1 Assessment of Societal and Cultural Issues

- Supports sustainability by encouraging rechargeable battery use and reducing waste.
- Aligns with cultural shifts toward energy-efficient, eco-friendly technologies.

5.5.2 Assessment of Health and Safety Issues

- Safety features (over-voltage, over-current, reverse polarity) protect users and equipment.
- Automatic cutoff prevents overcharging, reducing risks of overheating or failure.

5.5.3 Assessment of Legal Issues

- Adheres to safety and performance regulations for electrical devices.
- Avoids intellectual property violations by ensuring originality in design.

5.6 Sustainability Evaluation

The multifunctional battery charger promotes sustainability by supporting rechargeable batteries, reducing reliance on disposable batteries and minimizing environmental waste. Its energy-efficient design, featuring precise voltage regulation and automatic cutoff, conserves power and extends battery life.

Durable components reduce the need for frequent replacements, decreasing electronic waste. Additionally, compliance with environmental standards ensures responsible manufacturing and disposal practices. This design aligns with global efforts toward eco-friendly technology and sustainable energy solutions.

5.7 Ethical Issues

The development of the multifunctional battery charger raises several ethical considerations, primarily focusing on user safety, environmental impact, intellectual property, and fairness in design:

1. User Safety

- Ensuring that the charger operates safely, preventing risks such as overcharging, overheating, and electrical hazards, is a primary ethical responsibility. The charger is equipped with safety features like over-voltage, over-current, and reverse polarity protections, which are critical to safeguarding users and their devices from potential harm.
- Providing clear instructions and warnings for proper usage ensures that users are informed of potential risks, promoting responsible use of the product.

2. Environmental Responsibility

- The charger supports rechargeable batteries, helping reduce the environmental burden of disposable batteries. By promoting a sustainable approach, it encourages users to adopt energy-efficient solutions, which can contribute to reducing electronic waste (e-waste) and conserving resources.
- Ethical design also includes considering the life cycle of the charger itself, ensuring that materials used are recyclable and that the product is designed for durability, minimizing waste in the long term.

3. Intellectual Property

- The project ensures that all components, designs, and technologies used are original or properly licensed, avoiding infringement on patents or copyrighted materials. Respecting intellectual property rights upholds the ethical principle of fairness in engineering and innovation.

5.7 Individual Contribution of Each Member

Activities	Member Involved (ID)
Resource Collection	2006036,41,47,48
Simulation on Proteus	2006047,65
Rectifier Circuit	2006041,47
Pulse Circuit	2006036,41,60
Buck Circuit	2006048,60
Auto Cut-off Circuit	2006036,41,60
Auto Cut-Off Coding	2006065
Level Indicator Circuit	2006036,48,65
Presentation Slide	All

5.8 Mode of Team Work

The project was a result of effective teamwork, where tasks were divided based on individual strengths. Team members focused on different aspects, such as power electronics, microcontroller programming, and safety features. Open communication through regular meetings and digital platforms ensured smooth collaboration and problem-solving. Integration of components and testing was done collectively, with everyone contributing to identifying issues and refining the design. The final report and presentation were a joint effort, showcasing the collaborative nature of the project. Overall, teamwork, coordination, and shared goals were key to the project's success.

5.9 Diversity Statement of Team

Our team values diversity in all its forms, believing that different perspectives enhance creativity and problem-solving. We embraced diverse backgrounds, skills, and approaches, which contributed to innovative solutions and effective collaboration. Every team member's input was respected, fostering a productive and inclusive environment. This diversity was key to the success of the project.

5.10 Logbook of Project Implementation

Date	Milestone achieved	Individual Role	Team Role	Comments
20-09-2024	Resource collection	2006036,41,47,48		
24-09-2024	Project Proposal	All		Project Accepted
26-09-2024	Buying component	All		
02-10-2024	Proteus Simulation	2006047,65		Done successfully
05-11-2024	Project Update	All		
02-12-2024	Hardware (Rectifier Circuit)	2006041,47		
04-12-2024	Hardware (Pulse Circuit)	2006036,41,60		
05-12-2024	Hardware (Buck Circuit)	2006048,60		
05-12-2024	Hardware (Auto Cut-Off Circuit)	2006036,41,60		
07-12-2024	Auto Cut-Off Coding	2006065		
07-12-2024	Hardware (Level Indicator Circuit)	2006036,48,65		
10-12-2024	Final Testing	All		Working correctly
15-12-2024	Presentation Slide	All		

6 Project Management and Cost Analysis

6.1 Bill of Materials

Equipment	Unit	Price per unit	Price
Breadboard	05	100	500
Arduino Uno	01	650	650
Diode	15	02	30
LED	10	02	20
Delay	01	45	45
Resistor	12	05	60
Capacitor	06	15	90
Inductor	02	30	60

MOSFET	03	70	140
Pot	01	25	25
Wire box	02	150	300
Transformer	01	250	250
IC	02	20	40
PVC board			100
Total			2310

7 Future Work

- **Advanced Control Algorithms:** Implementing PID or adaptive control for better charging efficiency.
- **Wireless Charging:** Integrating wireless charging technology for added convenience.
- **IoT Integration:** Adding remote monitoring and control features through smartphones.
- **Expanded Battery Compatibility:** Supporting a wider range of battery types, including newer technologies.
- **Energy Efficiency:** Improving power conversion efficiency to reduce energy waste.
- **Enhanced Safety:** Adding advanced fault detection and thermal management.
- **Cost Reduction:** Focusing on reducing manufacturing costs for mass production.

8 References

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